

FORTUNA

Foundation List — Direction Funnel

Track A | v0.8 | 13 Jul 2026

A daily direction funnel — broadest macro scan at the top, narrowing to a single index-direction verdict, then branching into the two trading books. Every item keeps a permanent old-ID alias (MAC-1 was B4, ENG-11 was E10) so references match across every document.

v0.8 — the Foundation List trunk is complete at 22/22. This revision adds the three pre-market gates (PRE-1 India VIX, PRE-2 GIFT Nifty, PRE-5 calendar day-type), a new front-matter page on what the funnel actually outputs, the GIFT to PRE-3 opening handoff, and the post-Sep-2025 Tuesday-expiry weekday map.

What the Funnel Outputs

Read this first. The funnel does not output a **direction** — it outputs a **decision stack**. Direction is one field in it. On many days the most valuable output is *sit out*, or *this is a trap*, or *size down*. An indicator that says “don’t trade Wednesday” earns its keep as much as one that says “go long.” Not every item is about direction; each answers a different question.

The verdict object

The System Score should not be a single directional number. It should output a small **verdict object** — with direction as just one field, free to read *unknown* / *irrelevant* while the day is still fully classified:

```
verdict = { tradeable · day_type · direction · conviction · volatility_character · size · trap_risk · stop }
```

A · GO / NO-GO — should I trade at all today?

Output	What it tells you	Primary items
Trade / sit-out	Is today worth trading, or stand aside	PRE-1 VIX (>25 avoid), ENG-1 CHOP (>61.8), PRE-4 CPR wide, PRE-5 (Wed low-vol), ENG-8 liquidity
Day-type forecast	Trend / range / chop / dead day	PRE-4 CPR width, PRE-1 VIX, ENG-1, PRE-3 gap type, PRE-5 weekday
Event landmine	Is today dangerous (news / expiry)	MAC-1, NAT-1, PRE-5 (expiry / month-end)
Liquidity gate	Can I actually get filled	ENG-8 (index auto-pass; stock 4-metric screen)

B · CHARACTER — what kind of day / move is this?

Output	What it tells you	Primary items
Regime: trend vs chop	Trust the signals, or don’t	ENG-1 (ADX+CHOP) master · ENG-2 EMA slope
Volatility magnitude	How big are moves / how wide stops	ENG-9 ATR · PRE-1 VIX · PRE-4 CPR width · ENG-10 BB width
Volatility character	Clean expansion vs wave-like reversal chop that traps retail	PRE-1 VIX (high=whippy) · ENG-1 CHOP · MAC-1/NAT-1 (news) · PRE-3 exhaustion · ENG-8 divergence
Volume / participation	Is the move backed by real flow	ENG-8 (RVOL, breakout vol, climax, dry-up)

C · DIRECTION — which way?

Output	What it tells you	Primary items
Bias / direction	Up / down / sideways	ENG-2 EMA · ENG-3 RSI · ENG-4 MACD · ENG-5 Supertrend · +DI/–DI
Opening direction	Gap up / down at 9:15	PRE-2 GIFT · PRE-3 gap · MAC-2 cues

D · EXECUTION — where / when / how much / where's the exit?

Output	What it tells you	Primary items
Location (where)	Where entries trigger / moves stall	ENG-11 S/R+OI · ENG-7 VWAP · ENG-2 EMA · PRE-4 CPR levels
Timing (when)	The exact entry tick	ENG-6 StochRSI · ENG-7 VWAP reclaim · ENG-10 squeeze-fire · PRE-3 open range
Conviction	How much to trust the setup	ENG-1 (gates all) · ENG-8 volume · ENG-9 ATR · multi-TF alignment
Position sizing	How many lots	ENG-9 ATR (risk+stop) · PRE-1 VIX (inverse)
Risk / stop	Where's the exit	ENG-7 VWAP+hunt zone · ENG-5 Supertrend trail · ENG-9 ATR · ENG-11 structure
Premium / IV	Buy vs sell; IV-crush risk	PRE-1 VIX · IV percentile · NAT-1/MAC-1 (crush) · ENG-10 (buy cheap premium)

E · INTELLIGENCE — what's smart money doing / who's trapped?

Output	What it tells you	Primary items
Institutional intent	What FIIs / DIIs are positioned for	FLO-1 cash · FLO-2 derivatives L/S · ENG-11 OI walls
Trap / reversal warning	Retail being trapped; exhaustion	FLO-2 (FII-vs-retail divergence) · PRE-3 (exhaustion ~78% reverse) · ENG-8 (divergence/climax) · ENG-7 (failed reclaim) · ENG-3 (ceiling reject)
Time-of-day context	First-15-min noise · expiry-PM pin · last hour	PRE-3 · ENG-7 (first 30m) · OPT (max-pain PM) · PRE-5
Swing / positional backdrop	The multi-day / week regime	MAC-3 DXY/10Y · FLO trend · HTF ENG-2/3 · VIX extremes · NAT-1 elections

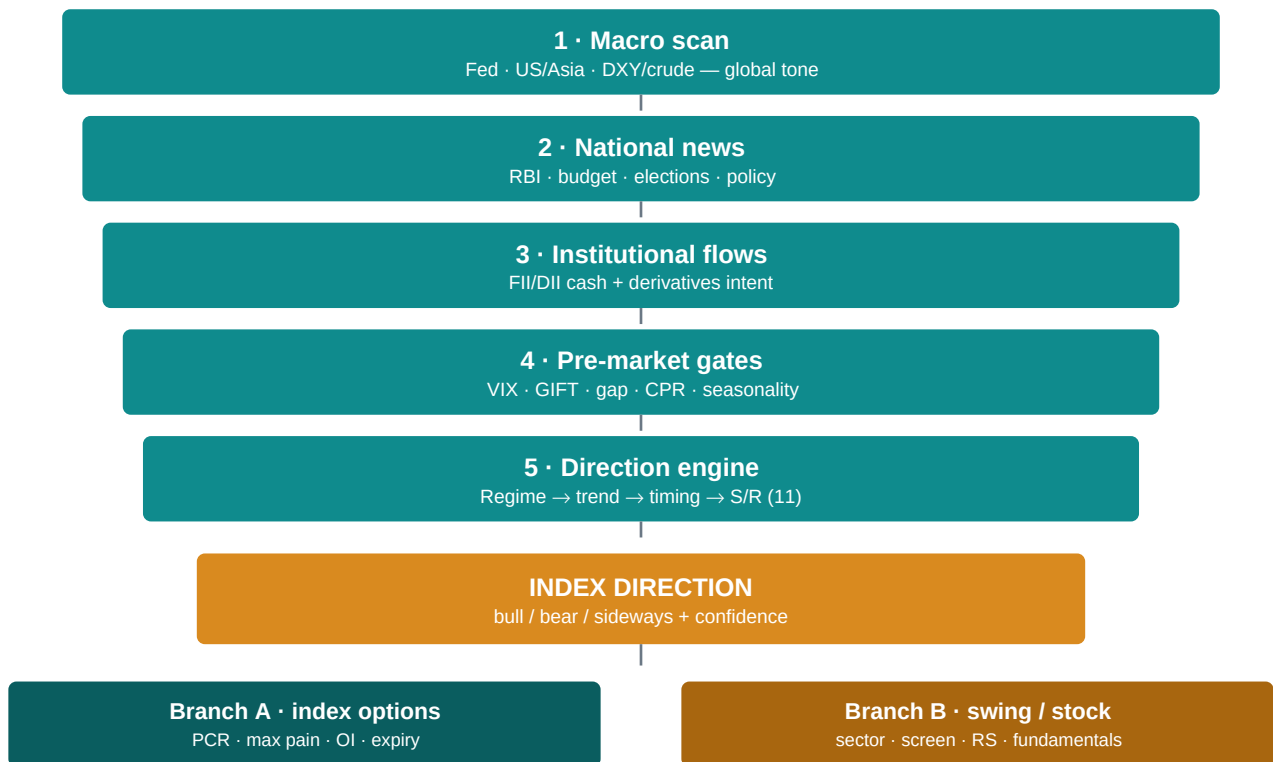
Two worked examples

Wednesday low-volume sit-out. A new day-type created by the Sep-2025 Tuesday-expiry shift — the opposite of folk wisdom, a fresh consequence of a rule change. Tuesday expiry concentrates volume/volatility into Tuesday; Wednesday is the post-expiry hangover (fresh weekly contract, unbuilt OI, thin volume, drift). Output = **reduced size / sit out**, a GO/NO-GO call, not a direction. Trust the mechanism; verify the magnitude on post-Sep-2025 data.

News-day wave-reversal trap. The clearest case of why direction-only framing fails. On these days direction is *unknowable* — price reverses in waves precisely to trap both sides. Correct output is not a side but **character = chaotic two-sided** → **sit out / wait for the rollover**. Produced by a cluster: high VIX (PRE-1) + high CHOP (ENG-1) + event flag (MAC-1/NAT-1) + failed VWAP reclaims (ENG-7) + volume divergence (ENG-8) + exhaustion gap (PRE-3). When the cluster lights up, the score outputs NO-TRADE / trap-risk.

The Direction Funnel

Macro → national → flows → pre-market → engine → index direction → two books



The teal stages are the shared trunk — read top-to-bottom every morning, they produce the index direction (also the Module I-8 dashboard order). The amber box is the single output. The two coloured boxes are the further filters per book: Branch A layers options intelligence onto the direction; Branch B narrows index → sector → stock.

Re-index Map

New funnel ID ↔ old research ID — use everywhere for consistent referencing.

New ID	Was	Item	Stage	Status
MAC-1	B4	Global macro news (Fed/geo/commodity)	Macro	DONE
MAC-2	B6	US / Asian market correlation	Macro	DONE
MAC-3	B12	Macro levels (DXY/US10Y/crude/gold)	Macro	DONE
NAT-1	B5	National news (RBI/budget/elections)	National	DONE
FLO-1	C1	FII/DII daily cash flow	Flows	DONE
FLO-2	C2	FII index derivatives positioning	Flows	DONE
PRE-1	G5	India VIX regime	Pre-market	DONE
PRE-2	G7	GIFT Nifty pre-market	Pre-market	DONE
PRE-3	E9	Gap up / gap down behaviour	Pre-market	DONE
PRE-4	D8-11	CPR (central pivot range)	Pre-market	DONE
PRE-5	A7	Day-of-week / seasonality	Pre-market	DONE
ENG-1	D8-7	Regime filter (ADX + Choppiness)	Engine	DONE
ENG-2	D8-6	EMAs across timeframes	Engine	DONE
ENG-3	D8-2	RSI (Cardwell ranges)	Engine	DONE
ENG-4	D8-1	MACD crossover	Engine	DONE
ENG-5	D8-8	Supertrend	Engine	DONE
ENG-6	D8-3	Stochastic RSI	Engine	DONE
ENG-7	D8-4	VWAP + SL decisions	Engine	DONE
ENG-8	D8-5	Volume & liquidity	Engine	DONE
ENG-9	D8-10	ATR (volatility backbone)	Engine	DONE
ENG-10	D8-9	Bollinger + squeeze	Engine	DONE
ENG-11	E10	Support & resistance + OI walls	Engine	DONE

All 22 trunk items are now researched. The three former pre-market gates (PRE-1 VIX, PRE-2 GIFT, PRE-5 seasonality) — previously mis-filed under Patterns / Backtesting — are completed in this revision. Next: the branch filters (OPT / STK), with stock-level work moving to Document B.

1

Macro scan

Global risk tone — the widest layer · MAC-1 · MAC-2 · MAC-3

MAC-1 — Global Macro News

was B4 · Fed · geopolitical · commodity shocks · Stage 1 · Macro

HEADLINE VERDICT

A real but conditional, mostly gap-delivered signal. Not continuously weighted — an event-gate / regime overlay: Neutral on ~90% of days, dominant on the rare event day. Edge is at the open and decays within 30–45 min. High volatility here is overnight gap risk, not in-session whipsaw.

Proof & mechanism

Fed rate news moves Indian equity/debt/forex via FII flows. Feb-2026 US–Israel–Iran corrected Nifty ~9%; resolution flipped FIIs to net buyers after 13 selling sessions. Crude: a \$110 spike hit Nifty same-day, but 16 spikes >20%/mo since 1999 gave positive 1–3mo returns in 13; corr ~0.36 — crude moves volatility more than direction.

Timeframe

Timezone means these are gap events. FOMC lands ~11:30 PM–2:30 AM IST and hits the 9:15 open; by ~9:45–10:00 it is priced.

Signal strength

Sub-signal	Direction	Volatility	Trade stance	Note
Fed / FOMC	MED	HIGH	CAUTION	Directional via FII, often pre-priced; overnight gap
Acute geopolitical	HIGH	V.HIGH	AVOID	Dominates while active; chaotic, wide gaps
Geo resolution	MED	HIGH	CAUTION	Mirror recovery; trade after confirmation
Crude shock (same-day)	MED	MED	CAUTION	Weak direction; mean-reverts; size down
Crude level / trend	LOW	LOW	AVOID	Excluded — owned by MAC-3

Per-timeframe weight

Chart	Weight	Rationale
Monthly	Very low	Structural resilience + crude mean-reversion.
Weekly	Low–mod	FII-flow response plays out over days when live.
Daily	Binary	~0 non-event; can dominate event days.
1H / intraday	High → 0 by 10:30	Sets gap + regime, then hand to VWAP.

FORMAL RECOMMENDATION

Include as a conditional event-gate / regime overlay, baseline Neutral. On event days caps confidence + flags two-sided risk; on acute shocks can override toward the gap. Fed + acute geopolitics are directional; crude = volatility/context with a mean-reversion guard.

MAC-2 — US / Asian Market Correlation

was B6 · GIFT · S&P;/Nasdaq · Nikkei/Hang Seng · Stage 1 · Macro

HEADLINE VERDICT

The most codeable macro item — a continuous, quantifiable correlation (Nifty–S&P; 30-day reached 0.68). Edge is at the open: cues set the gap + first 30 min, then decay. Trade/no-trade from cue alignment + VIX regime (aligned + VIX 12–16 = gap-and-go; contradiction / gap >200 / VIX>18 = fade/sit out).

Proof & mechanism

Bidirectional short-run causality with Dow/S&P; no long-run cointegration. NASDAQ daytime drives ~9.5% of Nifty overnight-return volatility. GIFT prices it all in real time; Nasdaq→IT is the sharpest sector channel.

Consumes PRE-2

The GIFT implied-gap sub-signal below is sourced from PRE-2 — MAC-2 uses it as one of three cues (US close + Asia + GIFT). Wire it once, reference it twice; do not double-count the gap.

Signal strength

Sub-signal	Direction	Volatility	Trade stance	Note
GIFT implied gap	HIGH	MED	TRADE	75–85% opening-direction when aligned, gap >100pts
US overnight close	HIGH	MED	CAUTION	Most important overnight input; Nasdaq→IT
Cue alignment (US+Asia+GIFT)	HIGH	LOW	TRADE	Gap-and-go conviction; VIX 12–16
Contradiction / gap>200 / VIX>18	LOW	HIGH	AVOID	Fade or sit out first 15 min
US trend regime (wk/mo)	MED	—	TRADE	0.68 co-move biases swing direction

Per-timeframe weight

Chart	Weight	Rationale
Monthly	Moderate	0.68 co-move biases trend; decoupling risk.
Weekly	Moderate	US swing trend co-moves; Fortuna-Cash input.
Daily	Mod–high	Sets the opening gap + risk tone every session.
1H / intraday	High at open → decays 10:00	Owens gap + first 30 min; Asia = live filter.

FORMAL RECOMMENDATION

Include as the opening-direction + swing-bias input: GIFT gap + US close + Asian confirmation, gated by VIX (12–16 = trade; >18/contradiction = fade; <50 pts = noise). Moderate weekly/monthly weight; decays by 10 AM. Route Nasdaq→IT to Fortuna-Cash.

MAC-3 — Macro Levels

was B12 · DXY · US 10Y · crude · gold · Stage 1 · Macro

HEADLINE VERDICT

The mirror image of the intraday items: strong at weekly/monthly, near-zero intraday. DXY yes (strong inverse); US 10Y yes (threshold-based); crude barely (defers to MAC-1); gold essentially no ($R^2 < 0.3\%$, risk-off confirm only). Route to swing + regime, not the intraday engine.

Proof & mechanism

Rising DXY → FII outflows → Nifty down (thresholds 100/106/110). US 10Y >4.5% coincides with FII selling; crossing 5% (Oct-23) sent Nifty -1,000 in a week. Gold flipped to +0.88 corr in 2021–25 — unreliable as a hedge now.

Signal strength

Sub-signal	Direction	Volatility	Trade stance	Note
DXY (dollar index)	MED	MED	CAUTION	Inverse, threshold-based, often leads
US 10Y yield	MED	MED	CAUTION	Non-linear; >4.5% pressure, 5% shock
Crude (level)	LOW	MED	CAUTION	Corr ~-0.36; defers shock to MAC-1
Gold	LOW	LOW	CAUTION	Not a predictor; regime confirm only

Per-timeframe weight

Chart	Weight	Rationale
Monthly	Mod-high (3–4/10)	DXY + 10Y are genuine positional drivers.
Weekly	Moderate	DXY/10Y trends bias swing; core Fortuna-Cash input.
Daily	Low	Only extreme threshold breaks matter same-day.
1H / intraday	Near-zero	Risk-on/off backdrop only.

FORMAL RECOMMENDATION

Include DXY + US 10Y as weekly/monthly directional inputs (threshold-banded) for swing + regime. Crude defers to MAC-1; drop gold as a predictor (risk-off confirm only). ~0 intraday.



2

National news

Domestic event flags · NAT-1

NAT-1 — National News

was B5 · RBI · Union Budget · elections · policy · Stage 2 · National

HEADLINE VERDICT

B5 events fire during NSE hours (RBI 10 AM, Budget 11 AM), not as overnight gaps — genuinely intraday-tradeable and dangerous, and they crush volatility after. For an options buyer the primary output is trade / no-trade + IV-crush warning, with bias only on a real consensus-surprise.

Proof

RBI: tone-driven (rate pre-priced). Budget: >1% move on 16/25 days, 2.65% avg intraday range, no reliable direction, VIX falls on all 15 days studied. Elections (surprise-only): 4-Jun-2024 −5.93% close (~■26 lakh cr wiped); expected 2019 moved just −0.69%. Regulatory (SEBI F&O;) = structural, not direction.

Signal strength

Sub-signal	Direction	Volatility	Trade stance	Note
Elections (results day)	V.HIGH	V.HIGH	AVOID	Surprise-only; VIX ramp + brutal post-crush
RBI MPC	MED	HIGH	CAUTION	Tone-driven; wait for 10 AM statement
Union Budget	LOW	V.HIGH	AVOID	No direction; whipsaw + VIX crush kills longs
Govt / regulatory policy	LOW	MED	CAUTION	Sector-specific; route to Fortuna-Cash

Per-timeframe weight

Chart	Weight	Rationale
Monthly	Low, exc. elections	Only a general-election surprise sets a multi-month tone.
Weekly	Mod on event weeks	RBI / Budget / election week; ~0 otherwise.
Daily	Binary	~0 normal; can dominate the event day.
1H / intraday	High post-announcement	Two-sided before; no-trade/caution flag pre-event.

FORMAL RECOMMENDATION

Include as a scheduled-event gate whose output for an options buyer is trade/no-trade + IV-crush warning, not a continuous weight. Bias only on a genuine surprise; default to caution + reduced size on all event days.

3

Institutional flows

Realized flow + leveraged intent · FLO-1 · FLO-2

FLO-1 — FII/DII Daily Cash Flow

was C1 · Net cash + 3–5 day rolling trend · Stage 3 · Flows

HEADLINE VERDICT

Cash data publishes after close (6–7 PM) — structurally a next-day / swing signal, never intraday. Real but moderate and regime-dependent (single-day $R^2 \sim 16.5\%$). The number that matters is the net (FII + DII), not FII alone — DII absorption prevents the 'FIIs selling so market falls' error.

Proof & mechanism

One of the most consistent relationships in Indian markets, but single-day explains $\sim 16.5\%$; 3+ sessions carry far more value. Self-reinforcing FII→rupee loop. FIIs momentum-trade, DIIs contrarian — so they offset.

Signal strength

Sub-signal	Direction	Volatility	Trade stance	Note
FII net cash (single day)	LOW	MED	CAUTION	Retrospective, EOD, noisy alone
FII 3–5 day rolling trend	MED	MED	TRADE	The reliable read; swing bias
DII net / absorption ratio	MED	LOW	CAUTION	The decoupling / floor detector
Sharp rupee fall + FII sell	MED	HIGH	AVOID	Compound risk-off

Per-timeframe weight

Chart	Weight	Rationale
Monthly	Moderate	Cumulative FII trend + structural DII floor.
Weekly	Mod–high	The 3–5 day rolling net flow is the sweet spot.
Daily	Mod (next-day)	Last evening's print biases the open.
1H / intraday	Low	Only yesterday's-flow context; no live signal.

FORMAL RECOMMENDATION

Include as a next-day / swing bias on the net (FII+DII) figure, 3–5 day rolling trend, and DII absorption ratio — normalized by turnover, regime-gated. ~ 0 same-day intraday. Never read FII alone. Pair with FLO-2.

FLO-2 — FII Index Derivatives Positioning

was C2 · Participant-wise OI · long/short ratio · Stage 3 · Flows

HEADLINE VERDICT

Correction to the old spec's 'intraday' note: participant-wise OI publishes EOD (~7 PM) — a next-day / multi-week signal (live intraday OI = the option chain, ENG-11). Its edge over FLO-1: it shows leveraged directional intent, and the long-short ratio leads reversals 24–48h.

Proof & mechanism

Sustained FII net-long index futures → markets rise; net-short → correct. Best read is divergence: retail buying calls at highs while FIIs liquidate longs + buy puts → market follows FIIs, traps retail. A confirmer, not a standalone trigger.

The hedge trap

An FII futures short is often a hedge vs a cash long, not a bear bet — always cross-check FLO-1 to tell bet from hedge.

Signal strength

Sub-signal	Direction	Volatility	Trade stance	Note
FII long-short ratio	MED	MED	TRADE	24–48h to multi-week lead
FII-vs-retail divergence	HIGH	MED	TRADE	The trap-reversal signal; best edge
FII net short (raw)	MED	HIGH	AVOID	Check FLO-1: short+cash-long = HEDGE
FII/Pro option writing	MED	LOW	CAUTION	S/R context; overlaps ENG-11

Per-timeframe weight

Chart	Weight	Rationale
Monthly	Moderate	Futures long-short trend frames the multi-week regime.
Weekly	Mod-high	C2's strongest horizon — leading positioning trend.
Daily	Mod (next-day)	Positioning + divergence bias the next open.
1H / intraday	Low	EOD data; live OI = ENG-11.

FORMAL RECOMMENDATION

Include as the institutional-intent layer — long-short ratio + FII-vs-retail divergence. Always read with FLO-1 to separate a directional short from a hedge; treat as a confirmer, not a trigger. Highest conviction when cash, futures and the option chain align.

4

Pre-market gates

The 9:00 AM scan — trade / no-trade + day-type · PRE-1..5

PRE-1 — India VIX Regime

was G5 · the master volatility gate · Stage 4 · Pre-market

HEADLINE VERDICT

The master volatility gate — the pre-market twin of ENG-1. **Zero directional weight:** every authoritative source states VIX does not predict up/down, only the expected size of moves. What it sets is trade/no-trade posture, position size, and — critically for an options buyer — how expensive premium is and how brutal the IV-crush risk is. Score it like ENG-1: it multiplies confidence and sets size; it never picks a side.

Does it affect direction?

No — the key correction. VIX-change → next-day gap direction is only ~53% (barely above chance). The relationship is inverse but asymmetric: on 7-Apr-2025 VIX surged 65% as Nifty fell 3.24%; on 28-Oct-2008 VIX fell only ~33% as Nifty rose 6.35% — a spike is a downside-risk flag, not a symmetric signal. The rigorous long-run correlation is -0.41 (the folk -0.85 is a stress-window figure). One real edge is positional, not intraday: extreme VIX mean-reverts and precedes rallies (post-2008 +80.8%, post-COVID +83.6%, post-Jul-2023 low +26.4%) — a Fortuna-Cash swing input.

Mechanism

VIX = aggregate implied vol of OTM Nifty options, annualised (15 ≈ ±4.33% over 30 days). The chain that matters: VIX level → premium richness → theta/IV-crush exposure → whether a directional buy can even pay off. Low VIX = cheap premium, slow grind; high VIX = rich premium, violent two-sided moves, IV-collapse risk on a correct call.

Signal strength

Sub-signal	Direction	Volatility	Trade stance	Note
VIX <13 (calm)	Gate/Size	LOW	TRADE	Cheap premium, grind-trend regime; buying viable
VIX 13–17 (normal)	Gate/Size	MED	TRADE	Workable zone; full cascade weight
VIX 17–20 (nervous)	Gate/Size	MED	CAUTION	Size down; premium richening
VIX >20 (elevated)	Gate/Size	HIGH	CAUTION	Buying penalised — IV-crush + wide stops
VIX >25 (turbulence)	Gate	V.HIGH	AVOID	Two-sided chaos; stand aside for buys
VIX delta (day/day)	—	—	AVOID	53% direction — do NOT trade the delta
Extreme reversion	Swing	HIGH	TRADE	Contrarian positional only — NOT intraday

Per-timeframe weight

Chart	Weight	Rationale
Monthly	Mod (positional)	Extreme-VIX mean-reversion is a genuine contrarian tell.
Weekly	Mod-high	Sets the week's premium-cost + gap-risk regime.
Daily	High (gate)	The pre-open trade/no-trade + sizing decision, both books.
1H / intraday	Gate (0 dir)	Conditions cascade confidence + stop width; 0 direction.

FORMAL RECOMMENDATION

Include as the master pre-market volatility gate feeding both books — the twin of ENG-1. Contributes 0 to direction; sets (a) trade/no-trade + size and (b) premium/IV-crush context. Use a rolling VIX percentile, not fixed bands (the regime drifted lower post-2023). Cross-confirm width with PRE-4: low VIX + narrow CPR = strongest trend day; high VIX + wide CPR = chop, sit out. Never trade the VIX delta.

PRE-2 — GIFT Nifty Pre-market

was G7 · the 6:30 AM opening-direction read · Stage 4 · Pre-market

HEADLINE VERDICT

The most directional of the pending gates — the one item where ‘does it predict direction?’ is the right question. GIFT is the USD Nifty futures contract on NSE IX (successor to SGX since Jul-2023), trading ~21h. It answers one thing well: which way and roughly how far the Nifty *opens* — not the day (like MAC-2). Its accuracy is partly mechanical, not predictive. Treat it as the gap forecaster that hands off to PRE-3, not a trigger.

Does it affect direction? (proof + skepticism)

Yes, for the opening print. Nifty spot is frozen overnight while GIFT trades on global news, so the widened basis *is* the implied gap; arbitrage pulls the open toward GIFT. The cited ~75–85% is a portal claim, not a backtest — plausible precisely because it is mechanical (forecasting the gap from indicators scores only ~53%). It measures opening direction only, and inflates by being mechanical. Validate your own hit-rate by gap-size bucket.

Mechanism & the handoff

Two sessions (~6:30 AM–3:40 PM, 4:35 PM–2:30 AM). Implied gap $\approx$ (GIFT – prev close) minus the standing carry basis (USD-INR rate differential; ~5–30 pts intraday, widening overnight). Cleanest read 8:45–9:10 AM. Feeds MAC-2 (as one cue) and PRE-3 (as the pre-print forecast). **GIFT red on real overnight news → high-confidence down OPEN; PRE-3 then classifies runaway vs exhaustion → enter 9:30–9:45, never the 9:15 print.**

Signal strength

Sub-signal	Direction	Volatility	Trade stance	Note
Large gap (>0.5%), US clear	HIGH	MED	TRADE	Strongest confirmation; open follows GIFT
Moderate gap (0.2–0.5%), no event	HIGH	MED	TRADE	Reliable opening lean; feed PRE-3
Small gap (<±0.2%)	LOW	LOW	AVOID	Noise band — mostly carry, no direction
Event day (RBI/Budget/expiry)	LOW	HIGH	AVOID	9:15 domestic flow overrides GIFT
Late reversal (6:30–9:15)	—	HIGH	CAUTION	Re-check at 9:00; early read can flip
Thin overnight liquidity	LOW	HIGH	CAUTION	Exaggerated move cash won't follow

Per-timeframe weight

Chart	Weight	Rationale
Monthly / Weekly	~0	No swing signal; a single session's open.
Daily	Mod (open only)	Sets the opening gap + first-30-min risk tone.
Pre-open (6:30–9:10)	High (its home)	The gap-direction + magnitude read.
Intraday after 9:15	~0	Hand off to PRE-3; GIFT is spent.

FORMAL RECOMMENDATION

Include as the pre-open gap forecaster — the 8:45–9:10 AM basis-adjusted read of direction + magnitude that feeds MAC-2 (cue) and PRE-3 (pre-print forecast). High weight in its window, zero after 9:15. Strong when large + news-driven; noise under ±0.2%; gate off on event days; refresh at 9:00 for flips. Reference-only (retail can't trade it). Validate the hit-rate by gap-size bucket rather than inheriting the folk 75–85%.

PRE-3 — Gap Up / Gap Down Behaviour

was E9 · opening classifier — the convergence point · Stage 4 · Pre-market

HEADLINE VERDICT

Gap direction alone is near a coin flip (Nifty gap-ups $\geq 1\%$ fill same-day only 29.4%). The edge is gap + trend-alignment + ATR-normalized size + VIX + no-news. This is where the whole macro/flow stack + VIX converges into the first trade of the day.

Four-type taxonomy

Common ($<0.5 \times \text{ATR}$, range) $\rightarrow$ fills $\sim 90\%$, noise. Runaway ($0.5\text{--}1.5 \times \text{ATR}$, with-trend, no news, VIX 12–16) $\rightarrow$ continues 55–65%. Breakaway (genuine catalyst) $\rightarrow$ trends, rarely fills. Exhaustion ($>1.5\text{--}2 \times \text{ATR}$, extended, climax vol, high VIX) $\rightarrow$ reverses $\sim 78\%$, gives back ~ 1 ATR.

Inverted-U + gates

Continuation peaks for moderate gaps, falls at both ends. Measure size $\div$ 14-day ATR. Gate by trend-alignment (ENG-2/3), no-news (MAC/NAT calendar), VIX (<20 go, >25 fade). Huge-gap reversals come later — wait for the rollover, don't fade at 9:15.

Signal strength

Sub-signal	Direction	Volatility	Trade stance	Note
Runaway: with-trend + no-news + VIX 12–16	MED	MED	TRADE	Break 9:30 H/L, with the gap; 55–65%
Breakaway: genuine catalyst	MED	HIGH	TRADE	Trends all day; rarely fills — don't fade
Exhaustion: huge + extended	MED	HIGH	CAUTION	$\sim 78\%$ reverse; fade after the rollover
Common: tiny / range	LOW	LOW	AVOID	Noise; spread eats options edge
Opening range (9:15–9:30)	MED	HIGH	CAUTION	First 15 min noise; enter 9:30–9:45

Per-timeframe weight

Chart	Weight	Rationale
Monthly / Weekly	Low	Breakaway/exhaustion classification for swing context.
Daily	Moderate	The open sets tone; gap type is a swing signal.
Intraday — first hour	High (its home)	The go/fill/sit-out decision.
After $\sim 11:30$	~ 0	78% of fills done; opening edge decayed.

FORMAL RECOMMENDATION

Include (strong, opening) as the opening classifier: four-type taxonomy via trend-alignment + ATR-normalized size + VIX + no-news. Runaway/breakaway $\rightarrow$ gap-and-go; exhaustion $\rightarrow$ fade after rollover; tiny/split $\rightarrow$ sit out. Never trade the first 15 min blindly.

PRE-4 — CPR (Central Pivot Range)

was D8-11 · pre-market day-type forecast + session S/R · Stage 4 · Pre-market

HEADLINE VERDICT

The Indian intraday staple. Three levels (BC/PP/TC) from the prior day's H/L/C — known before the open. CPR width forecasts the day type (narrow = trend day, wide = range day), which front-runs the coincident regime gate (ENG-1); the levels give an intraday S/R roadmap.

Width forecast

Narrow CPR (~<35 pts Nifty) = low vol yesterday → trending day; wide (~>60 pts) = range/chop day. Combine with VIX/CHOP: high VIX + narrow CPR = strong trend; low VIX + wide CPR = safe range day.

Levels

Above TC = bullish, below BC = bearish, inside BC–TC = indecision; PP = magnet. Virgin CPR reacts strongly on first test.

Signal strength

Sub-signal	Direction	Volatility	Trade stance	Note
Narrow CPR (day-type)	LOW	LOW	TRADE	Pre-market trend-day forecast; front-runs ENG-1
Wide CPR	LOW	HIGH	AVOID	Range-day forecast; mean-revert the levels
Price above TC / below BC	MED	MED	TRADE	Intraday directional bias
Virgin CPR tested	MED	MED	TRADE	Strong reaction on first test

Per-timeframe weight

Chart	Weight	Rationale
Intraday (daily-derived)	High	Morning session bias + day-type forecast.
Daily	High	Prior-day CPR governs today's session.
Weekly / Monthly CPR	Moderate	Swing S/R from prior week/month.
vs ENG-1	Complement	CPR = forward forecast; ADX/CHOP = coincident.

FORMAL RECOMMENDATION

Include (strong, intraday) as the morning roadmap: CPR width as a pre-market regime predictor (narrow = trend, wide = range) front-running ENG-1, plus the three levels as intraday S/R. Cross-confirm width with VIX/Choppiness.

PRE-5 — Calendar Day-Type / Seasonality

was A7 · weekday theta-cycle map + weak seasonal tilt · Stage 4 · Pre-market

HEADLINE VERDICT

Re-researched. The return-based seasonality (Monday-bullishness, month-of-year) is weak, contradictory and mostly insignificant — don't trade it. But a strong, in-scope output hides underneath: a **weekday day-type map** created by the Sep-2025 Tuesday-expiry regime, driven by the theta/gamma/OI cycle — deterministic math, so its shape is trustworthy even on <1 year of data. It sets volume, volatility character and theta pressure per weekday (GO/NO-GO, day-type, sizing, premium), never a side. (Full weekday map on the next page.)

Does it affect direction? (proof + skepticism)

Return-direction: no, still weak — day-of-week studies are sign-unstable and the recent rigorous work finds no significant large-cap effect; the ~58% Monday claim doesn't survive. But that was the wrong question. What the map affects reliably is *day-type and theta pressure*, which is deterministic (a 0-DTE Tuesday option pins and decays regardless of last year's returns). Near-zero directional weight, high context value.

Monthly / annual seasonality — the 'all seasons' answer

Researched and honestly weak. The literature is contradictory (Nov–Dec strength, Mar–May weakness, a January effect — all with different signs in different studies); the most recent rigorous Nifty50/BankNifty study (2026) finds only January significant, and *negative*, every other month insignificant → high efficiency. 'Sell in May' shows up globally including India but weak. For intraday index options this is essentially irrelevant — a months-long positional tilt. Note it as a faint Fortuna-Cash backdrop; contribute ~0 to the intraday score. Budget/earnings seasons are already owned by NAT-1/MAC-1 and STK-SEA (Doc B) — don't duplicate.

Signal strength

Sub-signal	Direction	Volatility	Trade stance	Note
Tue expiry (AM)	Context	HIGH	TRADE	High-vol/high-volume; range-break with volume
Tue expiry (PM)	Context	V.HIGH	AVOID	Pin + MM games + wide spreads; no new positions
Mon expiry-eve	Context	MED→HIGH	CAUTION	Theta headwind for buyers; seller-favoured
Wed post-expiry	Context	LOW*	AVOID	Low-volume drift; reduced size (verify)
Thu (BSE expiry)	Context	MED	TRADE	Buyer-friendlier; watch BSE vol spillover
Fri pre-weekend	Context	MED	TRADE	Fresh directional OK; no weak weekend holds
Monday return tilt	~0/noise	LOW	AVOID	Sign-unstable; don't trade
Month-of-year tilt	~0 intraday	—	AVOID	Positional, weak, out-of-scope

Per-timeframe weight

Chart	Weight	Rationale
Monthly	~0 (intraday)	Month-of-year is a weak months-long positional tilt; Fortuna-Cash backdrop only.
Weekly	Mod (day-type)	The weekday theta-cycle map — genuinely useful here.
Daily	Mod (context)	Sets day-type + theta/premium prior + GO/NO-GO.
Intraday	~0 (exc. Tue PM)	Only expiry-afternoon pin/theta, owned by OPT.

FORMAL RECOMMENDATION

Include (with caveats): strong as the weekday day-type / theta-cycle map — a mechanism-grounded, in-scope context gate outputting GO/NO-GO (Wed sit-out, Tue-PM avoid), day-type (Tue high-vol tradeable AM) and premium/theta prior (Mon buyer headwind) — never a direction. Drop the return-based day-of-week and month-of-year tilts as intraday signals; keep monthly seasonality only as a faint Fortuna-Cash backdrop. Validate every weekday magnitude on post-Sep-2025 data; trust the shape now.

The Weekday Day-Type Map — post-1-Sep-2025 (Nifty intraday)

Because Nifty weekly expiry is now **Tuesday**, the week's theta / OI / gamma rhythm shifted. This map is about volume, volatility character and theta pressure — not direction. The shape is deterministic (theta math); the magnitude of each cell needs your own post-Sep-2025 backtest.

Day	DTE	Character	Vol	Options-buyer stance
Mon	1	"Pressure cooker" — weekend news meets expiry-eve; Fri→Mon weekend theta compressed; vol building	Rising	Heavy theta headwind; seller-favoured; buy only strong confirmed directional
Tue	0 · EXPIRY	Highest volume (~+50%) + volatility; gamma peaks ATM; max-pain pin + MM games late; spreads widen after ~2 PM	Highest	AM tradeable (first-30-min break + volume); avoid new positions PM; read sharp closes as theta not trend
Wed	~6	Post-expiry hangover — fresh weekly contract, OI not yet built, drift (your observation)	Low*	Reduced size / sit out unless a clean trend prints; verify magnitude on new-regime data
Thu	~5	Nifty mid-cycle, more time value; Sensex expires on BSE today → possible vol spillover	Moderate	Directional buying viable (more DTE = lighter theta); watch BSE-driven spikes
Fri	~4	Pre-weekend; sellers initiate for 3-day weekend theta; weekend gap risk on holds	Moderate	Fine for fresh directional; don't carry weak longs over the weekend (theta + gap)

* Wednesday's low-volume drift is mechanism-plausible (fresh series, unbuilt OI) and matches your observation, but it's the one cell most needing your own backtest — the regime is ~10 months / ~40 Wednesdays old. Trust the shape, confirm the size.

Structural notes. Tuesday = Nifty pin day (NSE); Thursday = Sensex pin day (BSE) — two pressure points, not one. BankNifty is monthly-only (no weekly since Nov-2024) — most weeks it has no expiry pin, a cleaner intraday underlying. Weekend theta is now a 3-day Fri→Mon drain — punishing for Friday/Monday option-buyer holds. Holiday shifts move Tuesday expiry to the prior session — read the NSE circular, never assume the weekday.

5

Direction engine

Regime → trend → direction → location → timing → conviction → stop · ENG-1..11

ENG-1 — Regime Filter (ADX + Choppiness)

was D8-7 · the master gate — trade or chop? · Stage 5 · Engine

HEADLINE VERDICT

The master gate every other engine item assumed. It gives no direction — it decides whether the cascade's signals should be trusted at all. 0 directional weight; it multiplies every other item's confidence.

Two measures

ADX = trend strength (+DI/−DI = direction); Choppiness Index = trend existence/efficiency (directionless). Complementary — a trend can be strong AND choppy. Optional 3rd: Efficiency Ratio, 2-of-3 vote.

Codeable rule

CHOP<38.2 AND ADX>25 = trend → run the cascade at full weight. CHOP>61.8 = chop → stand aside / mean-revert. 38.2–61.8 = wait. Direction from +DI/−DI, never ADX/CHOP.

Signal strength

Sub-signal	Direction	Volatility	Trade stance	Note
CHOP<38.2 + ADX>25 (trend)	LOW	MED	TRADE	Green light: full weight to the cascade
CHOP>61.8 (chop)	LOW	LOW	AVOID	Red light: stand aside or mean-revert
ADX rising + DI separating	MED	MED	TRADE	Strengthening trend; DI gives the side
ADX<20 / falling	LOW	LOW	AVOID	No trend; momentum unreliable

Per-timeframe weight

Chart	Weight	Rationale
Every timeframe	Gate	Runs each TF; cascade wants trend-regime agreement top-down.
Higher-TF	Gate	Weekly/daily regime validates lower-TF signals.
Execution-TF	Gate	5m CHOP>61.8 inside a bull weekly → wait, don't force.

FORMAL RECOMMENDATION

Include as the master trend/range gate: CHOP + ADX (+ optional Efficiency Ratio) decides trend vs chop per TF; momentum gets full weight only in a trend. Contributes 0 to direction but multiplies every other item's confidence.

ENG-2 — EMAs Across Timeframes

was D8-6 · the structural backbone · Stage 5 · Engine

HEADLINE VERDICT

The backbone the cascade rides on. The EMA stack + slope per TF is the operational definition of that TF's trend. The 20/50/200 are the dynamic S/R where pullback entries bounce. Crossovers are lagging confirmation only.

Roles

9 = fast trigger; 20 = intraday anchor (shallow-pullback support); 50 = deeper pullback; 200 = regime filter. Bull stack: price>9>20>50, rising 200. Flat 200 = sideways, signals unreliable.

The entry setup

Pullback into the 20 EMA on declining volume, bounce with a volume spike + VWAP confluence = A setup. EMA = where, StochRSI = when, volume = conviction.

Signal strength

Sub-signal	Direction	Volatility	Trade stance	Note
EMA stack + slope (regime)	HIGH	MED	TRADE	Backbone — defines each TF's trend
Price vs rising/falling 200	HIGH	MED	TRADE	Major regime; long above, short below
Pullback to 20 EMA	MED	MED	TRADE	Dynamic support; shallow-dip bounce
Confluence (EMA+VWAP+S/R)	HIGH	LOW	TRADE	Highest-conviction entry location
Tangled / flat EMAs	LOW	MED	AVOID	Chop; no crossover saves you

Per-timeframe weight

Chart	Weight	Rationale
Monthly / Weekly	High	50/200 stack defines the dominant regime.
Daily	High	Swing structure + institutional 50/200 DMA.
4H / 1H	High	Intermediate structure + pullback zones.
15m / 5m	High	Intraday structure: 20 anchor, 9 trigger.

FORMAL RECOMMENDATION

Include (strong) as the regime + dynamic-S/R backbone: stack/slope defines each TF's trend (high weight every TF); 20/50/200 = pullback-entry locations; crossover = confirmation only. Use 9/20/50/200 primary; backtest 13-21-34 (A2).

ENG-3 — RSI (Cardwell Ranges)

was D8-2 · regime + leading shift · Stage 5 · Engine

HEADLINE VERDICT

Stronger and more leading than MACD — via Cardwell ranges, not naive 70/30. Bull 40–80 (40 floor), bear 20–60 (60 ceiling, ~55 on Nifty). The range-shift leads price; overbought pins in strong trends (ride). Ash's ceiling-rejection = exit, breakout = re-enter is exactly this framework.

Levels

Bull 40–80, super-bull 60–80+, bear 20–60, sideways 40–60 (avoid). 50 = divide; clean 50-cross in HTF direction after a pullback = continuation. Range shift: break 60 = bear→bull, breach 40 = bull→bear (leads price).

Validated

'Rejects 55 then breaks 2nd/3rd try' = range-shift through the ceiling. 'Stays 70–100 long' = super-trend momentum. 'Daily runs 3+ days' = range persistence.

Signal strength

Sub-signal	Direction	Volatility	Trade stance	Note
HTF range regime (W/D)	HIGH	MED	TRADE	Defines the whole bias — RSI's best use
Range shift (break 60/40)	HIGH	MED	TRADE	The '55 breakout'; leads price
Super-trend pinning	HIGH	MED	TRADE	Ride; don't fade
Ceiling rejection (55 reject)	MED	MED	CAUTION	Exit call, NOT buy put (asymmetric)
Sideways 40–60	LOW	MED	AVOID	Whipsaw; stand aside

Per-timeframe weight

Chart	Weight	Rationale
Monthly / Weekly	High	Sets the range regime governing all below.
Daily	High	3-day continuation + swing sweet spot.
4H / 1H	Mod-high	Intermediate regime + pullback-to-40.
15m / 5m	Moderate	50/55-cross timing in HTF direction.

FORMAL RECOMMENDATION

Include (strong) as a regime + leading-shift signal via Cardwell ranges. High weekly/daily weight. Encode the asymmetry: ceiling rejection = exit, breakout = re-enter, super-trend = ride, sideways = stand aside. Set range from HTF first; backtest the ceiling per instrument.

ENG-4 — MACD Crossover

was D8-1 · momentum confirm (lagging) · Stage 5 · Engine

HEADLINE VERDICT

A lagging, price-derived tool. Its value comes from the higher-TF MACD as a trend filter — the raw 5-min crossover is the weakest, most whipsaw-prone form. Demoted from 'primary' to confirmation, weighted by multi-TF agreement.

Mechanism

Three reads by increasing lag: histogram slope → signal-line cross → zero-line cross. Divergence is the one leading read. Works in trends, whipsaws in chop.

Caveat

On 5-min the 12/26 EMAs span 60–130 min — the cross fires after much of the move. Use the higher-TF MACD for bias; gate by ENG-1.

Signal strength

Sub-signal	Direction	Volatility	Trade stance	Note
Higher-TF trend filter	MED	MED	TRADE	Highest-value use for 5-min bias
Multi-TF alignment	MED	MED	TRADE	Confirmation stack; highest conviction
Zero-line + histogram slope	MED	MED	CAUTION	Regime + earliest momentum tell
Raw 5-min crossover	LOW	HIGH	AVOID	Lag + whipsaws; weakest form

Per-timeframe weight

Chart	Weight	Rationale
Monthly / Weekly	Low–mod	Regime, lags at turns.
Daily	Moderate	Cleanest MACD trend filter.
1H	Moderate	Best intraday trend filter.
5m	Low (confirm)	Whipsaw/lag; timing only after HTF agrees.

FORMAL RECOMMENDATION

Keep MACD as a higher-TF trend filter + confirmation, weighted by multi-TF agreement — never a standalone 5-min trigger. Prefer histogram slope + divergence over the lagging crossover.

ENG-5 — Supertrend

was D8-8 · trend filter + dynamic SL · Stage 5 · Engine

HEADLINE VERDICT

ATR-based trend line: green (buy) below price, red above, flips on a close through it. Volatility-adaptive + 'sticky', so it doubles as a dynamic trailing stop. India's most-used intraday indicator — but a clean trend filter, not an every-flip trigger (raw flips = 45–60% win).

Use as a filter

Only longs green, shorts red. Raw flips whipsaw in chop (3+ flips in 20 bars = stand flat). The line = a volatility-adaptive trailing SL (add 0.5×ATR buffer).

Settings + combo

Nifty ~7/3 or 10/2–3; BankNifty 10/3–3.5. Killer combo: Supertrend(10,2) + VWAP on Nifty 5m, buy only when green AND above VWAP (~65%+ follow-through).

Signal strength

Sub-signal	Direction	Volatility	Trade stance	Note
Regime (green/red filter)	MED	MED	TRADE	Only longs green, shorts red
The line as trailing SL	MED	MED	TRADE	Volatility-adaptive stop
Raw flip standalone	LOW	HIGH	AVOID	45–60% win; whipsaws in chop
Flip + VWAP + HTF agree	MED	MED	TRADE	The filtered combo; ~65%+

Per-timeframe weight

Chart	Weight	Rationale
Weekly / Daily	Moderate	Higher-TF filter for the 15m signal.
15m	Mod (primary)	Primary Indian intraday TF for direction.
5m	Mod (timing)	Entry timing after 15m confirms.
SL role	All TFs	The line = a trailing stop.

FORMAL RECOMMENDATION

Include as a trend filter + dynamic trailing SL, gated by ENG-1 and confirmed by VWAP + higher-TF agreement — never an every-flip entry. Turn off 30 min before RBI/Budget/FOMC.

ENG-6 — Stochastic RSI

was D8-3 · entry-timing trigger (fastest) · Stage 5 · Engine

HEADLINE VERDICT

The fastest but noisiest oscillator — a pure entry-timing trigger, not direction. It fires at the pullback bottom the RSI/MACD cascade predicted, and whipsaws the most, so it's gated hardest. RSI decides which way; StochRSI decides the exact tick.

Mechanism

'Momentum of momentum' — the real signal is %K crossing %D inside an extreme zone ($OB > 0.8 / OS < 0.2$); a zone-touch without the crossover is noise.

Caveat

Pins at extremes worse than RSI — never fade the pinning. In a bull, only oversold crosses = re-entries; overbought crosses = exit/ignore, not puts. 14-3-3 (21-5-5 to slow).

Signal strength

Sub-signal	Direction	Volatility	Trade stance	Note
%K/%D out of OS (bull)	MED	MED	TRADE	Times the pullback bottom — CALL entry
%K/%D out of OB (bear)	MED	MED	TRADE	Mirror; only with a bear cascade
OB cross in a bull	LOW	MED	CAUTION	Exit/ignore, NOT buy put
Pinned at extreme	MED	MED	TRADE	Embedded momentum; ride
Chop crosses	LOW	MED	AVOID	Worst whipsaw of the three

Per-timeframe weight

Chart	Weight	Rationale
Monthly / Weekly	Low	Too fast for regime; RSI/MACD own HTF.
Daily	Low–mod	Times daily swing entries (21-5-5).
4H / 1H	Moderate	Pullback timing within the trend.
15m / 5m	Mod–high	Its home — the entry trigger.

FORMAL RECOMMENDATION

Include as the fast entry trigger at the bottom of the cascade — %K/%D cross out of oversold (bull)/overbought (bear) = the 5m/15m tick. Gate hardest; counter-trend crosses = exit not reversal; never fade the pinning.

ENG-7 — VWAP + SL Decisions

was D8-4 · intraday anchor + SL hub · Stage 5 · Engine

HEADLINE VERDICT

The intraday complement to EMA — institutional fair-value line. Above = call bias, below = put, flat-oscillating = no-trade. Pullback to VWAP + 20 EMA = A+ entry. Bands are NOT S/R (price walks them in trends). Also the hub of the four-SL framework.

Reclaim / rejection

Lose VWAP, sellers fail on rising volume, reclaim = bullish continuation. Rally into VWAP from below, roll = flip to resistance. Break + failed retest kills most false signals (your G8 hunt, cross-referenced).

Four-SL framework

VWAP SL, Hunt Zone = VWAP – 0.5×ATR (stay-in buffer), Structure SL (deeper = exit), ATR SL (size). One close below = watch; two = hunt; reclaim = stay; deeper than hunt = structure SL.

Signal strength

Sub-signal	Direction	Volatility	Trade stance	Note
Above/below VWAP (bias)	HIGH	MED	TRADE	Core intraday call/put lean
Pullback to VWAP + 20 EMA	HIGH	LOW	TRADE	Highest-conviction entry location
VWAP reclaim on volume	MED	MED	TRADE	Chop-to-trend trigger
Rejection / failed reclaim	MED	MED	TRADE	Support→resistance flip
Oscillating flat VWAP	LOW	MED	AVOID	Range-day signature; no edge

Per-timeframe weight

Chart	Weight	Rationale
Monthly / Weekly	Low (anchored)	Session VWAP resets; use anchored from swings.
Daily	Low–mod	Anchored VWAP as multi-day S/R.
1H / 15m	High	Intraday bias + S/R + pullback zone.
5m	High	Execution anchor: reclaim/rejection + hunt-zone SL.

FORMAL RECOMMENDATION

Include (strong, intraday) as the bias anchor + dynamic S/R + SL hub. Pullback + 20 EMA = A+ entry; reclaim on volume = trigger. SL: VWAP SL + hunt zone → structure SL. Bands = context only, never mean-revert in a trend. First 30 min unreliable; index needs a futures-volume proxy.

ENG-8 — Volume & Liquidity

was D8-5 · conviction filter + tradeability gate · Stage 5 · Engine

HEADLINE VERDICT

Two signals under one name, neither a direction-setter: (1) volume-as-confirmation = a conviction multiplier on the cascade's direction; (2) liquidity-as-gate = a binary tradeability filter. Nifty/BankNifty pass automatically; stock options must clear a 4-metric liquidity screen or spreads kill the edge.

Confirmation

Breakout needs volume >50% over 20-day avg (else fakeout); divergence = exhaustion; climax (3–10×) = blow-off; dry-up precedes breakout. Low volume ≠ no move (ease-of-movement).

Liquidity gate

Illiquid stock options cost 5×+ the spread; check volume + OI + spread + bid/ask SIZE. Index has no native volume → futures proxy. Spreads widen with VIX.

Signal strength

Sub-signal	Direction	Volatility	Trade stance	Note
Breakout volume expansion	MED	MED	TRADE	Validates a cascade-confirmed breakout
RVOL intraday	MED	MED	TRADE	Live participation gauge
Volume divergence	LOW	MED	CAUTION	Move on fumes; exhaustion
Volume climax	LOW	HIGH	CAUTION	Take profit; don't chase
Liquidity gate	LOW	MED	AVOID	Stock options must pass; index auto-pass

Per-timeframe weight

Chart	Weight	Rationale
Monthly / Weekly	Low–mod	Confirms major breakouts.
Daily	Mod–high	Breakout-volume for swing conviction.
15m / 5m	Moderate	RVOL + trigger-candle volume.
Liquidity gate	Binary	Per-instrument, always; hard GO/NO-GO.

FORMAL RECOMMENDATION

Include as (1) a conviction multiplier on the cascade's direction and (2) a hard liquidity gate — stock options must clear a 4-metric screen. Never a direction signal. Index needs a futures-volume proxy.

ENG-9 — ATR (Average True Range)

was D8-10 · volatility backbone · Stage 5 · Engine

HEADLINE VERDICT

The volatility backbone, not a direction signal. Feeds Supertrend bands, Keltner/TTM, SL sizing, the hunt zone (VWAP – $0.5 \times \text{ATR}$) and position sizing — the engine behind the Volatility column. 0 directional weight, essential plumbing.

What it signals

ATR expansion = volatility rising = move starting; contraction = calm (low percentile = squeeze setup). ATR spike (bar range $> \text{ATR}$) = momentum igniting. $\text{SL} = 1.5 \times \text{ATR}$; lots = risk $\div$ ATR-stop.

Per-instrument

Nifty, BankNifty, mid-caps have very different ATRs — all ATR-derived levels must be per-instrument (BankNifty's larger ATR = higher Supertrend multiplier).

Signal strength

Sub-signal	Direction	Volatility	Trade stance	Note
ATR expanding + trend	MED	HIGH	TRADE	'The move is real'
ATR spike	MED	HIGH	TRADE	Momentum start (with trend)
ATR contracting	LOW	LOW	CAUTION	Compression → squeeze setup
High ATR + chop	LOW	HIGH	AVOID	Wide whipsaws; wide stops

Per-timeframe weight

Chart	Weight	Rationale
Every timeframe	Infra	Volatility per TF; feeds SL/sizing + regime.
Directional weight	Zero	Substrate for Supertrend/Bollinger/SL.
Per-instrument	Required	Nifty vs BankNifty vs stocks differ.

FORMAL RECOMMENDATION

Include as the volatility-regime + SL/sizing backbone (0 directional weight). Track ATR + percentile per instrument per TF; use for all SL distances ($1.5 \times \text{ATR}$), the hunt zone, and sizing.

ENG-10 — Bollinger Bands + Squeeze

was D8-9 · volatility-expansion timer · Stage 5 · Engine

HEADLINE VERDICT

Keep the squeeze, not band-touch mean-reversion (fails in trends). The squeeze = bands compress = coiled spring = a big directional move is loading — exactly what an options buyer wants. The squeeze tells you when, not which way.

Squeeze mechanics

BandWidth <~4% = squeeze; longer squeezes resolve more violently. TTM Squeeze: BB(20,2) inside Keltner(20,1.5×ATR) = ON; expand back outside = FIRED = trigger; momentum histogram gives direction (~68–72% filtered).

For options

Buy premium DURING the squeeze (low IV), profit on expansion. Nifty squeezes before expiry/events — pairs with NAT-1 (buy the squeeze early, not into the event). Require volume to avoid head-fakes.

Signal strength

Sub-signal	Direction	Volatility	Trade stance	Note
Squeeze ON	LOW	LOW	CAUTION	Prepare: buy cheap premium, move loading
Squeeze FIRED + histogram	MED	HIGH	TRADE	Direction from histogram + volume
Band-touch in range	LOW	MED	CAUTION	Mean-revert only if confirmed range
Band-touch in trend	LOW	MED	AVOID	Price walks the band; don't fade

Per-timeframe weight

Chart	Weight	Rationale
Monthly / Weekly	Moderate	Daily/weekly squeeze = swing move loading.
Daily	Moderate	Classic squeeze-to-breakout horizon.
15m / 5m	Moderate	Intraday burst; pre-event compression.
Direction	—	Squeeze = when; histogram = which way.

FORMAL RECOMMENDATION

Include as the squeeze / volatility-expansion detector (the 'when is a move coming' tool) — NOT band-touch mean-reversion. TTM Squeeze + histogram; direction from histogram/volume; require volume to avoid head-fakes.

ENG-11 — Support & Resistance + OI Walls

was E10 · the 'where' map · Stage 5 · Engine

HEADLINE VERDICT

Not a direction signal — the location layer defining where entries trigger, moves stall, and pullbacks bottom. Its entire edge is confluence (a lonely line is near-worthless). Works because it's self-fulfilling — everyone watches the same levels, so stops pile just beyond them (your G8 hunt).

Types — price + OI

Price: prior swing H/L, day/week/month H/L/C, round numbers, 52w H/L + dynamic EMA/VWAP/CPR/Fib. Options-OI (the edge): Call Wall (resistance), Put Wall (support), the OI range, Max Pain (expiry magnet), Change-in-OI (reinforced = holds, abandoned = breaks). Writer-first lens: high Call OI = writers betting price stays below.

Break vs bounce

Bounce (wait for the reaction + volume), breakout (close + volume + follow-through), retest (role reversal, highest-probability). Zones not lines. Auto-flag prices where ≥ 3 level-types stack = A+ zones.

Signal strength

Sub-signal	Direction	Volatility	Trade stance	Note
Confluence zone (3+ types)	HIGH	MED	TRADE	Prior H/L + round + EMA/VWAP/CPR + OI wall
Call wall / Put wall	MED	MED	TRADE	Writer-defended intraday S/R
Change-in-OI	MED	MED	CAUTION	Reinforced holds; abandoned breaks fast
Role reversal (flip)	MED	MED	TRADE	Broken $S \leftrightarrow R$; very reliable
Max pain (expiry)	LOW	LOW	CAUTION	Near expiry only; trend day overrides

Per-timeframe weight

Chart	Weight	Rationale
Monthly / Weekly	High	Major levels frame the regime; strongest S/R.
Daily	High	Prior-day H/L/C, swing levels, 50/200 DMA.
4H / 1H	Mod-high	Intermediate structure; intraday zones.
Options-OI (walls/max pain)	High (intraday)	Intraday range + expiry gravity; resets each expiry.

FORMAL RECOMMENDATION

Include (strong) as the 'where' map: price-chart S/R + an auto-confluence detector (≥ 3 types = A+ zone), plus options-OI S/R (call/put walls, max pain, change-in-OI, writer-first lens). Confluence > lonely level; wait for the reaction not the touch; OI resets each expiry.

Direction Sequence — Coding Pipeline

How the engine items chain (new IDs) — gate-first, top-down. Each stage can veto the next. Stages 1–2 gate everything; each higher timeframe (M→W→D→4H→1H→15m→5m) gates the one below.

#	Stage	Question	Items	Gate / action
1	Regime gate	Trend or chop?	ENG-1 ADX+CHOP · PRE-4 CPR width	Chop / wide CPR → STOP. Runs first.
2	Higher-TF bias	Which way?	ENG-2 EMA · ENG-3 RSI (M→4H)	Sets dominant direction. Conflicting → wait.
3	Direction confirm	Momentum agrees?	ENG-4 MACD · ENG-5 Supertrend	Must agree with step 2, else no trade.
4	Location (where)	Pullback bottom?	ENG-11 S/R+OI · ENG-7 VWAP · ENG-2 EMA	Confluence zone (3+ types) = entry area.
5	Timing (when)	Turn happening?	ENG-6 StochRSI	%K/%D out of OS/OB at the zone = trigger.
6	Conviction	Move real?	ENG-8 Volume · ENG-9 ATR · ENG-10 squeeze	Weak vol / no expansion → skip / size down.
7	Risk (stop)	Where's the exit?	ENG-7 VWAP · ENG-5 Supertrend · ENG-9 ATR	SL from VWAP / Supertrend / structure ± ATR.

THE RULE. regime → bias → direction → location → timing → conviction → stop. Pre-open, PRE-3 (gap) sets the first-hour bias. A bull cascade buys CALLS on the with-trend trigger; the counter-trend signal is an exit / wait, never a PUT (mirror in a bear cascade).

Pre-open handoff — GIFT (PRE-2) → gap classifier (PRE-3)

GIFT gets you leaning the right way overnight; PRE-3 keeps you from buying the exhaustion-gap bottom. The sequence:

Step	Read	Action
1 • Overnight	GIFT red on real, decisive news (Dow >0.5%, clear catalyst); discount >0.5%	Lean short; high-confidence Nifty OPENS down
2 • Magnitude	Basis-adjusted implied gap; is it >0.5% or noise (<±0.2%)?	Noise → stand down; large → proceed
3 • 9:00 refresh	Any 6:30–9:15 reversal? Offsetting domestic event? VIX already blown?	Flip / gate off if so
4 • Open (9:15)	Gap prints; DO NOT trade the print	Hand to PRE-3
5 • Classify	PRE-3: runaway/breakaway (continue) vs exhaustion (~78% reverse)	Runaway → with-trend; exhaustion → wait for rollover
6 • Enter	9:30–9:45 on confirmed type + volume; PRE-1 IV-crush check	Buy puts after confirmation, not at 9:15

'Same way' is reliable for the OPEN's *direction*, not the session or the magnitude. A fear-driven gap-down is often an exhaustion gap that opens down then rallies — and high overnight VIX can IV-crush a correctly-directional put. Read PRE-1 + PRE-2 + PRE-3 together.

The Two Books — Branch Filters

Once the funnel outputs index direction, each book applies its own further filters. Document A owns Branch A (index options); Branch B (swing / stock) moves to Document B, which consumes Document A's index direction as its first filter rather than re-deriving it.

Branch	Filters (in order)	Output
A · Index options (Nifty/BankNifty intraday) — this document	Options intelligence (PCR, max pain, OI buildup, IV percentile, Greeks) → expiry behaviour (Tue weekly, last-Tue monthly) → 5-min execution cascade + VWAP SL-hunt	Strike + CE/PE
B · Swing / stock (individual name) — Document B	Inherit index direction (1st filter) → sector performance & rotation → NSE / Nifty-500 screen → relative strength vs Nifty → fundamentals (results, NISM Ch.8) → stock technical cascade (same engine, per-stock params) → STK-SEA stock/sector seasonality → stock-option liquidity gate	Stock swing / stock option

Two-document architecture. The shared trunk (MAC/NAT/FLO/PRE/ENG) lives in Document A and is *referenced* by Document B, not duplicated — one source of truth. Document B reuses the ENG cascade with per-stock parameters (different ATR, RSI ceilings, liquidity checks) and starts from 'inherit index direction from Document A'. Same funnel philosophy, same ID convention (STK- prefix, old-alias kept), so the two read as one system. Stock/sector seasonality (earnings windows, annual sector cycles) lives in Document B as STK-SEA — not here.

Foundation List Tracker

Funnel order · new IDs · direction / volatility rollup. Y = include, Y* = include with caveats. Gate / Infra / Fcast / Context = non-directional roles.

ID	Was	Item	Stage	Dir	Vol	Incl.
MAC-1	B4	Global macro news	Macro	HIGH	HIGH	Y*
MAC-2	B6	US / Asian correlation	Macro	HIGH	MED	Y
MAC-3	B12	Macro levels (DXY/10Y/crude/gold)	Macro	MED	MED	Y*
NAT-1	B5	National news	National	V.HIGH	V.HIGH	Y*
FLO-1	C1	FII/DII cash flow	Flows	MED	MED	Y*
FLO-2	C2	FII derivatives positioning	Flows	MED	MED	Y
PRE-1	G5	India VIX regime	Pre-mkt	Gate	Gate	Y
PRE-2	G7	GIFT Nifty	Pre-mkt	HIGH	MED	Y
PRE-3	E9	Gap behaviour	Pre-mkt	MED	MED	Y
PRE-4	D8-11	CPR (central pivot range)	Pre-mkt	HIGH	Fcast	Y
PRE-5	A7	Calendar day-type / seasonality	Pre-mkt	Context	—	Y*
ENG-1	D8-7	Regime filter (ADX+CHOP)	Engine	Gate	Gate	Y
ENG-2	D8-6	EMAs across timeframes	Engine	HIGH	MED	Y
ENG-3	D8-2	RSI (Cardwell ranges)	Engine	HIGH	MED	Y
ENG-4	D8-1	MACD crossover	Engine	MED	MED	Y*
ENG-5	D8-8	Supertrend	Engine	MED	MED	Y*
ENG-6	D8-3	Stochastic RSI	Engine	MED	HIGH	Y
ENG-7	D8-4	VWAP + SL decisions	Engine	HIGH	MED	Y
ENG-8	D8-5	Volume & liquidity	Engine	MED	MED	Y
ENG-9	D8-10	ATR	Engine	Infra	HIGH	Y
ENG-10	D8-9	Bollinger + squeeze	Engine	MED	HIGH	Y
ENG-11	E10	Support & resistance + OI	Engine	HIGH	MED	Y

22 of 22 funnel items researched — the Track A trunk is complete. The three former pre-market gates are now written up: PRE-1 (VIX, master gate), PRE-2 (GIFT, opening-only), PRE-5 (calendar day-type). Next: branch filters — Branch A (OPT: PCR / max pain / OI / expiry) here; Branch B (STK) in Document B. Backtesting A1–A7 and remaining patterns are Track C.

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